

$$C(S, t) = N(d_1)S - N(d_2)Ke^{-rT}$$

CALL OPTION
PRICE

STOCK
PRICE

CUMULATIVE
DISTRIBUTION
FUNCTION

STRIKE
PRICE

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$T = (T_1 - t)$ ----- TIME LEFT TILL MATURITY (IN YEARS)

r ----- RISK FREE RATE

σ ----- VOLATILITY

This model is based on 6 main variables that are used to establish the fair price of an option contract. It also has a formula which may be very intimidating at first, but we won't be using it as such. We will focus on the 6 variables that are used and understand the model through them. We will understand how each factor affects the pricing. Based on that we will understand what makes the price of an option contract move.